

Risk and Risk Factors

100 Questions: Algorithmic Bias, Fairness, Explainability, GenAI Risks

GARP RAI Exam Preparation

QUESTION: Why is understanding AI risks particularly challenging?

OPTIONS:

- ① AI systems are always perfectly reliable
- ② AI is hugely pervasive in range and variety, with rapidly developing capabilities
- ③ AI risks are all well-understood and documented
- ④ AI systems never fail

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** AI is used across many domains (finance, healthcare, justice, etc.), and AI capabilities are developing faster than our understanding of risks. This creates challenges for impact prediction, risk measurement, and governance.
- **Why A is wrong:** AI systems are NOT always reliable - they can and do fail
- **Why C is wrong:** Many AI risks are still being discovered and understood
- **Why D is wrong:** AI systems can fail, as demonstrated by numerous real-world examples

EXAM TRAP: AI risks are **evolving, pervasive, and high-stakes** - not static or fully understood!

QUESTION: Why does the pervasiveness of AI pose a risk management challenge?

OPTIONS:

- ① AI is only used in one industry
- ② Risks take different forms depending on the use case
- ③ All AI risks are identical
- ④ Pervasiveness makes risks easier to predict

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** AI applications span many domains including credit scoring, healthcare, policing, and employment. Each domain has unique risk profiles - a loan denial algorithm has different risks than a medical diagnostic algorithm.
- **Why A is wrong:** AI is used across MANY industries, not just one
- **Why C is wrong:** Risks vary significantly by use case
- **Why D is wrong:** Pervasiveness makes risks HARDER to predict, not easier

EXAM TRAP: AI risks are **context-dependent** across different use cases - no one-size-fits-all approach!

Q3: High-Stakes AI Use - Tutorial

QUESTION: What is a consequence of deploying AI in high-stakes settings?

OPTIONS:

- ❶ Technological shortcomings have minimal impact
- ❷ Technological shortcomings can have significant consequences for people
- ❸ AI is never used in high-stakes settings
- ❹ High-stakes use eliminates all risks

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Loan applications affect financial well-being, medical diagnoses affect health outcomes, bail decisions affect personal freedom. Algorithmic errors in these domains can have profound impacts on people's lives.
- **Why A is wrong:** High-stakes means impacts are SIGNIFICANT, not minimal
- **Why C is wrong:** AI IS used in high-stakes settings across many domains
- **Why D is wrong:** High-stakes use increases, not eliminates, risk

EXAM TRAP: High-stakes AI = significant consequences for individuals - this is why risk management is critical!

QUESTION: What challenge does rapid AI development create?

OPTIONS:

- ❶ It makes risk measurement easier
- ❷ It creates challenges for impact prediction, risk measurement, and governance
- ❸ It eliminates all regulatory concerns
- ❹ It slows down innovation

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** AI capabilities advance faster than our understanding of risks. Regulatory responses lag behind technology. Governance mechanisms are difficult to establish quickly. This allows potentially harmful practices to persist.
- **Why A is wrong:** Rapid development makes measurement HARDER
- **Why C is wrong:** Regulatory concerns are significant and growing
- **Why D is wrong:** Rapid development ACCELERATES innovation, it doesn't slow it

EXAM TRAP: Rapid AI development = **governance and regulation challenges** - regulation can't keep pace with technology!

QUESTION: How are AI risks typically related?

OPTIONS:

- ① They are completely independent of each other
- ② They are often interconnected - lack of transparency affects fairness, bias leads to reputational risks
- ③ There is only one type of AI risk
- ④ Risks are never interconnected

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** AI risks form a complex network. Lack of transparency makes it impossible to assess fairness. Algorithmic bias leads to reputational damage. Safety failures can reveal fairness issues. Risks reinforce each other.
- **Why A is wrong:** Risks are highly interconnected
- **Why C is wrong:** There are many types of AI risks (functional, data, organizational, strategic, societal)
- **Why D is wrong:** Risks ARE interconnected - they don't exist in isolation

EXAM TRAP: AI risks are **interconnected and mutually reinforcing** - need holistic approach!

QUESTION: What is algorithmic bias?

OPTIONS:

- ① Random errors in algorithmic predictions
- ② A systematic deviation of an algorithm's output, performance, or impact relative to some norm or standard
- ③ Algorithms that are always perfectly accurate
- ④ Errors that only occur in supervised learning

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Algorithmic bias is systematic (not random), can affect output, performance, or impact, and is relative to some norm, aim, standard, or baseline. It can be explicit (intentional) or implicit (unintended).
- **Why A is wrong:** Bias is SYSTEMATIC, not random - it follows patterns
- **Why C is wrong:** Algorithms CAN be biased - no algorithm is always accurate
- **Why D is wrong:** Bias can occur in ANY type of learning (supervised, unsupervised, reinforcement)

EXAM TRAP: Algorithmic bias = systematic deviation from a norm - not random errors!

Q7: Explicit vs Implicit Bias - Tutorial

QUESTION: What is the difference between explicit and implicit bias?

OPTIONS:

- ① Explicit bias is always worse than implicit bias
- ② Explicit bias is intentional/projected by developers; implicit bias creeps in undetected
- ③ Implicit bias is intentional; explicit bias is accidental
- ④ There is no difference

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Explicit bias involves the developer intentionally projecting their own biases into the algorithm. Implicit bias creeps in undetected through design choices or training data. Both can lead to unfair outcomes.
- **Why A is wrong:** Both types can be equally harmful to affected individuals
- **Why C is wrong:** This is reversed - explicit is intentional, implicit is unintended
- **Why D is wrong:** They differ in terms of intentionality

EXAM TRAP: Explicit = **intentional**; Implicit = **unintended** - both can cause harm!

Q8: Individual Fairness Definition - Tutorial

QUESTION: What is individual fairness in algorithmic design?

OPTIONS:

- ① Treating all groups equally
- ② The doctrine of "treating like cases alike"
- ③ Ensuring equal outcomes across demographics
- ④ Always using protected characteristics

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Individual fairness is based on the Aristotelian doctrine that similarly situated individuals should be treated equally. It focuses on individuals, not groups, and requires agreement on what counts as "alike."
- **Why A is wrong:** This is group fairness, not individual fairness
- **Why C is wrong:** This is demographic parity (a group fairness measure)
- **Why D is wrong:** Individual fairness typically AVOIDS using protected characteristics

EXAM TRAP: Individual fairness = [treating like cases alike](#) - focus on individuals, not groups!

QUESTION: What is group fairness?

OPTIONS:

- ① Treating each individual uniquely
- ② Examining statistical differences across groups
- ③ Only considering individuals with protected characteristics
- ④ Ignoring group-level patterns

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Group fairness looks at statistical differences across groups sharing protected characteristics. It asks whether certain groups are disadvantaged by the algorithm (e.g., admission rates by gender, loan approvals by race).
- **Why A is wrong:** This is individual fairness
- **Why C is wrong:** Group fairness considers ALL groups, not just those with protected characteristics
- **Why D is wrong:** Group fairness EXAMINES group patterns - that's the whole point

EXAM TRAP: Group fairness = **statistical differences across groups** - looks at group-level patterns!

QUESTION: In the college admission example, what fairness issue arises?

OPTIONS:

- ① All applicants are treated fairly
- ② Individual fairness requires agreeing on what counts as "alike"
- ③ Socioeconomic factors never matter
- ④ Algorithms always make perfect decisions

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** What counts as "alike" is often contentious. Should we consider hardships overcome? How to translate resilience into numbers? Different people have different views on what criteria matter.
- **Why A is wrong:** Fairness is not automatically achieved - it requires careful consideration
- **Why C is wrong:** Socioeconomic factors ARE often considered in admission decisions
- **Why D is wrong:** Algorithms can make unfair decisions

EXAM TRAP: Individual fairness requires **agreement on what is "alike"** - which is often contested!

Q11: Demographic Parity Definition - Tutorial

QUESTION: What is demographic parity?

OPTIONS:

- ① Predictions must be equally accurate across groups
- ② The distribution of predictions must be identical across subpopulations
- ③ Positive cases must be correctly identified equally across groups
- ④ Precision must be equal across groups

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Demographic parity requires $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$. The distribution of predictions is identical across groups, ensuring the same outcome rate (e.g., same admission rate across genders).
- **Why A is wrong:** This is predictive rate parity/equal opportunity
- **Why C is wrong:** This is equal opportunity
- **Why D is wrong:** This is predictive rate parity

EXAM TRAP: Demographic parity = **equal distribution of predictions** - independent of correctness!

Q12: Demographic Parity Formula - Tutorial

QUESTION: What is the mathematical definition of demographic parity?

OPTIONS:

- ① $P(Y = 1 | \hat{Y} = 1, A = 0) = P(Y = 1 | \hat{Y} = 1, A = 1)$
- ② $P(\hat{Y} = 1 | A = 0) = P(\hat{Y} = 1 | A = 1)$
- ③ $P(\hat{Y} = 1 | Y = 1, A = 0) = P(\hat{Y} = 1 | Y = 1, A = 1)$
- ④ $P(Y = 1 | A = 0) = P(Y = 1 | A = 1)$

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** This is the correct formula where $\hat{Y}$ is the predicted outcome and A is the protected attribute. It states that the probability of a positive prediction is equal regardless of group membership.
- **Why A is wrong:** This is predictive rate parity (uses precision)
- **Why C is wrong:** This is equal opportunity (uses sensitivity)
- **Why D is wrong:** This is base rate equality - a different concept

EXAM TRAP: Demographic parity = $P(\hat{Y} = 1 | A = 0) = P(\hat{Y} = 1 | A = 1)$ - predictions, not outcomes!

Q13: Demographic Parity Drawback - Tutorial

QUESTION: What is a major drawback of demographic parity?

OPTIONS:

- ① It is always impossible to achieve
- ② It does not consider the quality/accuracy of predictions
- ③ It is too computationally expensive
- ④ It requires protected characteristics

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Demographic parity is independent of true values. It can be satisfied even if all predictions are wrong (e.g., unqualified candidates labeled as qualified equally across groups). It doesn't ensure predictions are correct.
- **Why A is wrong:** Not always impossible, but can be challenging when base rates differ
- **Why C is wrong:** It's computationally simple - that's part of its appeal
- **Why D is wrong:** It does require knowing group membership, which may not always be available

EXAM TRAP: Demographic parity ignores **prediction accuracy/quality** - a significant limitation!

Q14: Predictive Rate Parity Definition - Tutorial

QUESTION: What is predictive rate parity?

OPTIONS:

- ① The distribution of predictions is identical across groups
- ② Precision ($TP/(TP+FP)$) is equal across groups
- ③ Sensitivity ($TP/(TP+FN)$) is equal across groups
- ④ Accuracy is equal across groups

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Predictive rate parity requires $P(Y = 1 | \hat{Y} = 1, A = 0) = P(Y = 1 | \hat{Y} = 1, A = 1)$. Among those predicted positive, the proportion correctly identified is the same across groups (equal precision).
- **Why A is wrong:** This is demographic parity
- **Why C is wrong:** This is equal opportunity
- **Why D is wrong:** This is accuracy parity - a different concept

EXAM TRAP: Predictive rate parity = equal precision across groups - "of predicted positives, how many were right?"

QUESTION: What is a drawback of predictive rate parity?

OPTIONS:

- ① It ignores true positives
- ② It does not account for differences in base rates across groups
- ③ It is always impossible to achieve
- ④ It requires no data

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Base rates may differ significantly across groups. Achieving predictive rate parity becomes extremely challenging when base rates differ, and can have adverse effects on algorithmic performance (may require sacrificing accuracy).
- **Why A is wrong:** It specifically considers true positives through precision
- **Why C is wrong:** Not always impossible, but can be difficult
- **Why D is wrong:** Requires data to calculate precision

EXAM TRAP: Predictive rate parity struggles with **different base rates** - creates trade-offs!

QUESTION: What is equal opportunity?

OPTIONS:

- ① The distribution of predictions is identical across groups
- ② Precision is equal across groups
- ③ Sensitivity ($TP/(TP+FN)$) is equal across groups
- ④ All predictions are always correct

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Equal opportunity requires $P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$. The true positive rate (sensitivity) is equal across groups, ensuring positive cases are equally likely to be correctly identified.
- **Why A is wrong:** This is demographic parity
- **Why B is wrong:** This is predictive rate parity
- **Why D is wrong:** No model is always correct - this is impossible

EXAM TRAP: Equal opportunity = **equal true positive rate (sensitivity)** - critical in medical diagnostics!

QUESTION: What is the mathematical definition of equal opportunity?

OPTIONS:

- ① $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$
- ② $P(Y = 1|\hat{Y} = 1, A = 0) = P(Y = 1|\hat{Y} = 1, A = 1)$
- ③ $P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$
- ④ $P(Y = 1|A = 0) = P(Y = 1|A = 1)$

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** This formula states that the probability of a correct positive prediction ($\hat{Y} = 1$) given the actual positive outcome ($Y = 1$) is equal across groups ($A=0$ and $A=1$).
- **Why A is wrong:** This is demographic parity
- **Why B is wrong:** This is predictive rate parity
- **Why D is wrong:** This is base rate equality

EXAM TRAP: Equal opportunity = $P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$ - equal true positive rate!

QUESTION: What is equalized odds?

OPTIONS:

- ① Equal distribution of predictions
- ② Equal precision across groups
- ③ Equal true positive rate AND equal false positive rate across groups
- ④ Equal accuracy across groups

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Equalized odds is more restrictive than equal opportunity. It requires both sensitivity equality ($P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$) AND specificity equality ($P(\hat{Y} = 1|Y = 0, A = 0) = P(\hat{Y} = 1|Y = 0, A = 1)$).
- **Why A is wrong:** This is demographic parity
- **Why B is wrong:** This is predictive rate parity
- **Why D is wrong:** Accuracy parity is a different concept

EXAM TRAP: Equalized odds = equal TPR AND equal FPR - most stringent fairness measure!

QUESTION: What does the fairness impossibility theorem state?

OPTIONS:

- ❶ All fairness measures can be satisfied simultaneously
- ❷ It is impossible to satisfy demographic parity, predictive rate parity, and equal opportunity simultaneously when base rates differ
- ❸ Fairness is always achievable
- ❹ Only one fairness measure exists

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** When base rates differ between populations, it is mathematically impossible to satisfy all three fairness measures (demographic parity, predictive rate parity, and equal opportunity) simultaneously. Trade-offs are unavoidable.
- **Why A is wrong:** This is impossible - the theorem proves it
- **Why C is wrong:** Fairness has inherent trade-offs
- **Why D is wrong:** There are multiple fairness measures

EXAM TRAP: Cannot satisfy **all fairness measures** simultaneously when base rates differ - must prioritize!

QUESTION: Why is accuracy not always a good performance measure?

OPTIONS:

- ① It is always a perfect measure
- ② It can be misleading with unbalanced data
- ③ It is too difficult to calculate
- ④ It only works for regression

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** With unbalanced data, high accuracy can be misleading. Example: 100 applicants, 10 suitable, 90 unsuitable - algorithm classifies all as unsuitable, accuracy = 90
- **Why A is wrong:** Accuracy has significant limitations
- **Why C is wrong:** Accuracy is easy to calculate
- **Why D is wrong:** Accuracy is for classification, not regression

EXAM TRAP: Accuracy is **misleading with unbalanced data** - always check class distribution!

QUESTION: What is precision in classification?

OPTIONS:

- ① $TP/(TP+FN)$
- ② $TP/(TP+FP)$
- ③ $(TP+TN)/(TP+TN+FP+FN)$
- ④ $TN/(TN+FP)$

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Precision = $TP/(TP+FP)$. It measures the proportion of positive predictions that are correct. Also called Positive Predictive Value (PPV).
- **Why A is wrong:** This is recall/sensitivity ($TP/(TP+FN)$)
- **Why C is wrong:** This is accuracy
- **Why D is wrong:** This is specificity

EXAM TRAP: Precision = $TP/(TP+FP)$ - "of predicted positives, how many were right?"

QUESTION: What is sensitivity (recall) in classification?

OPTIONS:

- ① $TP/(TP+FN)$
- ② $TP/(TP+FP)$
- ③ $(TP+TN)/(TP+TN+FP+FN)$
- ④ $TN/(TN+FP)$

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** Sensitivity = $TP/(TP+FN)$. It measures the proportion of actual positives that are correctly identified. Also called True Positive Rate (TPR) or Recall.
- **Why B is wrong:** This is precision
- **Why C is wrong:** This is accuracy
- **Why D is wrong:** This is specificity

EXAM TRAP: Sensitivity = $TP/(TP+FN)$ - "of actual positives, how many did we catch?"

QUESTION: What trade-off exists between precision and sensitivity?

OPTIONS:

- ① They always move in the same direction
- ② There is often a trade-off; maximizing one can reduce the other
- ③ They are completely independent
- ④ Precision is always more important

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** To catch all positives (high sensitivity), you may increase false positives (lower precision). To avoid false positives (high precision), you may miss positives (lower sensitivity). This is a fundamental trade-off.
- **Why A is wrong:** They often move in opposite directions
- **Why C is wrong:** They are related through the decision threshold
- **Why D is wrong:** Importance depends on context

EXAM TRAP: Precision and sensitivity have a **trade-off** - choosing the right balance depends on context!

QUESTION: What is the False Positive Rate (FPR)?

OPTIONS:

- ① $FP/(TP+FP)$
- ② $FP/(FP+TN)$
- ③ $FN/(TP+FN)$
- ④ $TN/(TN+FP)$

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** $FPR = FP/(FP+TN)$. It measures the proportion of actual negatives that are incorrectly classified as positive. Also equals $1 - \text{Specificity}$.
- **Why A is wrong:** This is $1 - \text{precision}$, not FPR
- **Why C is wrong:** This is False Negative Rate
- **Why D is wrong:** This is specificity

EXAM TRAP: FPR = $FP/(FP+TN)$ - important for ROC curves and fairness analysis!

QUESTION: What are the four components of a confusion matrix?

OPTIONS:

- ① TP, TN, FP, FN
- ② Mean, Median, Mode, Standard Deviation
- ③ Training, Validation, Test, Production
- ④ Precision, Recall, Accuracy, F1

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** TP (True Positive), TN (True Negative), FP (False Positive), FN (False Negative) are the four components that form the confusion matrix, which is the foundation for classification metrics.
- **Why B is wrong:** These are statistical measures, not confusion matrix components
- **Why C is wrong:** These are data splits, not confusion matrix components
- **Why D is wrong:** These are performance metrics derived from the confusion matrix

EXAM TRAP: Confusion matrix = TP, TN, FP, FN - foundation for all classification metrics!

QUESTION: What question does demographic parity answer?

OPTIONS:

- ① "Is the likelihood of a true positive prediction the same across groups?"
- ② "Is the likelihood of a positive prediction the same across groups?"
- ③ "Is the likelihood of identifying actual positives as positive the same across groups?"
- ④ "Are all predictions accurate?"

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Demographic parity asks whether the probability of a positive prediction ($\hat{Y} = 1$) is the same regardless of group membership. It focuses on equal outcome distribution.
- **Why A is wrong:** This is predictive rate parity
- **Why C is wrong:** This is equal opportunity
- **Why D is wrong:** This is accuracy

EXAM TRAP: Demographic parity = **equal positive prediction probability** - independent of correctness!

QUESTION: What question does equal opportunity answer?

OPTIONS:

- ① "Is the likelihood of a positive prediction the same across groups?"
- ② "Is the likelihood of a true positive prediction the same across groups?"
- ③ "Is the likelihood of identifying actual positives as positive the same across groups?"
- ④ "Is the likelihood of a negative prediction the same across groups?"

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Equal opportunity asks whether the true positive rate (sensitivity) is the same across groups - whether actual positives are equally likely to be correctly identified regardless of group membership.
- **Why A is wrong:** This is demographic parity
- **Why B is wrong:** This is predictive rate parity
- **Why D is wrong:** This is not a standard fairness measure

EXAM TRAP: Equal opportunity = **equal true positive rate** - "of actual positives, how many did we catch?"

QUESTION: What is a base rate?

OPTIONS:

- ① The error rate of an algorithm
- ② The general prevalence of something within a population
- ③ The accuracy of predictions
- ④ The precision of predictions

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Base rate refers to the general prevalence of something within a population. For example, the prevalence of a disease, or the proportion of defaulters. Base rates can differ significantly across groups.
- **Why A is wrong:** Error rate is different from base rate
- **Why C is wrong:** Accuracy is different from base rate
- **Why D is wrong:** Precision is different from base rate

EXAM TRAP: Base rate = prevalence in a population - key for understanding fairness trade-offs!

QUESTION: Why is balancing performance and fairness challenging?

OPTIONS:

- ① Fairness always improves performance
- ② Optimizing for fairness can degrade performance and vice versa
- ③ They are always perfectly aligned
- ④ Performance is never affected by fairness

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** There is often a fundamental trade-off between performance and fairness. Fairness constraints may reduce overall accuracy, and optimizing for performance may create unfair outcomes.
- **Why A is wrong:** Fairness often TRADES OFF with performance
- **Why C is wrong:** They are often in tension, not aligned
- **Why D is wrong:** Performance IS affected by fairness constraints

EXAM TRAP: There is often a [trade-off between performance and fairness](#) - must choose based on context!

QUESTION: Which fairness measure is most appropriate for medical diagnostics?

OPTIONS:

- ① Demographic parity
- ② Predictive rate parity
- ③ Equal opportunity
- ④ Accuracy parity

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Equal opportunity ensures that positive cases (patients with disease) are equally likely to be caught across groups. Missing a disease (false negative) can be life-threatening, so equal sensitivity is critical.
- **Why A is wrong:** Demographic parity ignores prediction accuracy - not appropriate for medical diagnostics
- **Why B is wrong:** Predictive rate parity focuses on positive predictions, not on catching disease
- **Why D is wrong:** Accuracy may be misleading with imbalanced data

EXAM TRAP: Medical diagnostics prioritizes **equal opportunity** (don't miss disease) - context determines the right measure!

QUESTION: Why can problem specification introduce bias?

OPTIONS:

- ❶ It always produces perfect algorithms
- ❷ How a problem is specified affects algorithmic performance and fairness
- ❸ Problem specification never affects outcomes
- ❹ All problems are specified identically

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Choice of criteria affects who applies and who gets hired. Translation to measurable features requires proxies. Some concepts are harder to quantify than others (e.g., productivity vs. teamwork).
- **Why A is wrong:** Problem specification can introduce significant bias
- **Why C is wrong:** Problem specification significantly affects outcomes
- **Why D is wrong:** Problems vary widely in how they are specified

EXAM TRAP: Problem specification is **critical for fairness** - how you define the problem matters!

QUESTION: What is a proxy variable in algorithmic design?

OPTIONS:

- ① A variable that perfectly measures the desired concept
- ② A variable used as a stand-in for a concept that is hard to measure directly
- ③ A variable that is never used in algorithms
- ④ A variable with no correlation to the desired concept

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Proxies are used when direct measurement is difficult (e.g., productivity measured by sales). They reduce complex concepts to simpler measurable constructs but risk introducing bias if imperfect.
- **Why A is wrong:** Proxies are imperfect by definition
- **Why C is wrong:** Proxies ARE used in algorithms
- **Why D is wrong:** Proxies should be correlated with the concept

EXAM TRAP: Proxies are **imperfect stand-ins** for hard-to-measure concepts - can introduce bias!

Q33: Amazon Recruiting Tool Example - Tutorial

QUESTION: What happened with Amazon's recruiting tool?

OPTIONS:

- ① It worked perfectly and was deployed
- ② It systematically downgraded CVs from women applicants
- ③ It eliminated all bias
- ④ It was immediately successful

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Amazon attempted to create an unbiased recruiting tool but the algorithm systematically downgraded women's CVs. Even identical CVs were rated lower for women. Amazon eventually abandoned the project.
- **Why A is wrong:** The tool failed due to bias
- **Why C is wrong:** It perpetuated existing biases
- **Why D is wrong:** The project was abandoned

EXAM TRAP: Amazon's tool **systematically discriminated against women** - demonstrates difficulty of fair AI!

QUESTION: What is "fairness through unawareness"?

OPTIONS:

- ① Using all available features
- ② Removing protected characteristics from input features
- ③ Always including protected characteristics
- ④ Ignoring all features

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Fairness through unawareness is the intuitive approach of removing protected characteristics from input features, assuming this prevents discrimination. However, correlated variables can still carry protected information.
- **Why A is wrong:** This would increase bias risk
- **Why C is wrong:** This would allow explicit discrimination
- **Why D is wrong:** This would make predictions impossible

EXAM TRAP: Fairness through unawareness **does NOT eliminate bias** - correlated variables still carry protected information!

QUESTION: In the FTA prediction example, what happened when protected characteristics were removed?

OPTIONS:

- ❶ Bias was completely eliminated
- ❷ The coefficient on prior convictions increased from 54
- ❸ The algorithm became perfectly fair
- ❹ No changes occurred

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The protected variable and prior convictions were correlated. Removing the protected variable contaminated the prior variable, increasing its coefficient from 54
- **Why A is wrong:** Bias was NOT eliminated
- **Why C is wrong:** The algorithm remained biased
- **Why D is wrong:** Significant changes occurred

EXAM TRAP: Removing protected characteristics **can make bias worse** through correlated variables!

QUESTION: What is historical bias in AI?

OPTIONS:

- ❶ Bias that occurs only in the present
- ❷ Bias in training data that reflects past discrimination
- ❸ Bias that never affects algorithms
- ❹ Bias that is always intentional

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Supervised learning algorithms learn from historical data, which reflects past biases and discrimination. If women were historically less likely selected for certain roles, the algorithm learns and may amplify this pattern.
- **Why A is wrong:** Historical bias is about the PAST
- **Why C is wrong:** Historical bias DOES affect algorithms
- **Why D is wrong:** Historical bias is often unintentional

EXAM TRAP: Historical bias = [past discrimination reflected in training data](#) - algorithms learn from the past!

QUESTION: What is a feedback loop in AI systems?

OPTIONS:

- ❶ A system that corrects its own errors
- ❷ When algorithmic predictions influence future data, perpetuating bias
- ❸ A system that never changes
- ❹ A system that eliminates all bias

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Feedback loops occur when algorithmic predictions influence real-world decisions, which create new data that reflects those biased decisions. The algorithm learns from this biased data, amplifying the original bias.
- **Why A is wrong:** Feedback loops often AMPLIFY errors
- **Why C is wrong:** Feedback loops cause change (often negative)
- **Why D is wrong:** Feedback loops can perpetuate bias

EXAM TRAP: Feedback loops **amplify and perpetuate bias** - become self-reinforcing!

QUESTION: What is representation bias?

OPTIONS:

- ① When data is perfectly representative
- ② When the collected sample is not representative of the target population
- ③ When all groups are equally represented
- ④ When data collection is unbiased

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Representation bias occurs when the collected sample does not reflect the target population. Some groups are over-represented, others under-represented, leading to problematic model specifications.
- **Why A is wrong:** Representation bias means NOT representative
- **Why C is wrong:** This would be balanced representation
- **Why D is wrong:** Representation bias IS a form of bias

EXAM TRAP: Representation bias = **non-representative sample** - algorithms perform better on dominant groups!

QUESTION: What is measurement bias?

OPTIONS:

- ① When measurement methods are consistent across the sample
- ② When measurement methods are not consistent across the sample
- ③ When all measurements are perfectly accurate
- ④ When measurements are always reliable

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Measurement bias occurs when measurement methods are not consistent across the sample. Different groups are measured differently, which can favor certain groups and create biased data.
- **Why A is wrong:** Measurement bias means INCONSISTENT methods
- **Why C is wrong:** Measurement bias means NOT perfectly accurate
- **Why D is wrong:** Measurement bias means UNRELIABLE across groups

EXAM TRAP: Measurement bias = inconsistent measurement across groups - creates systematic differences!

QUESTION: What is data composition bias?

OPTIONS:

- ① When datasets are perfectly balanced
- ② When minority subgroups are underrepresented, causing algorithms to underperform
- ③ When all groups are equally represented
- ④ When data is always complete

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Data composition bias occurs when minority subgroups are underrepresented in the data. The algorithm underperforms for these groups because there are fewer data points to learn from.
- **Why A is wrong:** Composition bias means IMBALANCED data
- **Why C is wrong:** This would be balanced data
- **Why D is wrong:** Composition bias means INCOMPLETE representation

EXAM TRAP: Data composition bias = underrepresented minority subgroups - algorithms underperform for these groups!

QUESTION: What does balancing a dataset mean?

OPTIONS:

- ① Using proportional representation
- ② Ensuring relevant minority subgroups occupy a similar proportion as majority groups
- ③ Removing all minority data
- ④ Always using the original distribution

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Balancing means increasing the representation of minority groups to ensure the algorithm recognizes them as statistically relevant. This is different from proportional representation.
- **Why A is wrong:** Proportional representation is different from balancing
- **Why C is wrong:** Balancing INCREASES minority representation
- **Why D is wrong:** Balancing changes the original distribution

EXAM TRAP: Balancing = **over-representing minorities** for algorithm performance - not proportional representation!

QUESTION: What are methods for balancing datasets?

OPTIONS:

- ① Over-sampling minority groups
- ② Under-sampling majority groups
- ③ Both A and B
- ④ Only removing outliers

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** Both over-sampling (adding more data from minority groups) and under-sampling (removing data from majority groups) are methods for balancing datasets. They can be used separately or together.
- **Why A is wrong:** Correct but incomplete
- **Why B is wrong:** Correct but incomplete
- **Why D is wrong:** Outlier removal is a different technique

EXAM TRAP: Bootstrapping methods include **over-sampling AND under-sampling** - often used together!

QUESTION: What is intersectional bias?

OPTIONS:

- ❶ Bias that only affects one group
- ❷ Bias affecting groups at the intersection of multiple underrepresented subgroups
- ❸ Bias that never occurs
- ❹ Bias that is always intentional

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Intersectional bias affects groups at the intersection of multiple underrepresented subgroups (e.g., women of color in facial recognition). Performance is particularly low for these intersectional groups.
- **Why A is wrong:** Intersectional bias affects MULTIPLE groups
- **Why C is wrong:** Intersectional bias DOES occur
- **Why D is wrong:** Intersectional bias is often unintentional

EXAM TRAP: Intersectional bias = **multiple underrepresented dimensions combined** - hardest to detect and correct!

QUESTION: How can word embeddings introduce bias?

OPTIONS:

- ① They always produce unbiased results
- ② They can contain stereotypes (e.g., man:computer programmer as woman:homemaker)
- ③ They never contain stereotypes
- ④ They are always perfectly neutral

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Word embeddings capture meaning from text, which contains societal stereotypes. Research has shown that embeddings can reinforce stereotypes (e.g., "man is to computer programmer as woman is to homemaker").
- **Why A is wrong:** Word embeddings CAN contain bias
- **Why C is wrong:** Word embeddings DO contain stereotypes
- **Why D is wrong:** Word embeddings reflect societal biases

EXAM TRAP: Word embeddings can contain **harmful stereotypes** - reflect societal biases in text!

QUESTION: How can objective functions introduce bias?

OPTIONS:

- ① They never affect fairness
- ② What the algorithm optimizes for makes an ethically significant difference
- ③ All objective functions are equally fair
- ④ Objective functions always eliminate bias

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The choice of objective function (what the algorithm optimizes for) has ethical significance. Different objectives (performance vs. error distribution) have different fairness implications depending on context.
- **Why A is wrong:** Objective functions significantly affect fairness
- **Why C is wrong:** Different objectives have different fairness implications
- **Why D is wrong:** Objective functions CAN introduce bias

EXAM TRAP: Objective function choice = **ethical significance** - context determines appropriate objective!

QUESTION: In the cancer detection example, why might different algorithms require different approaches?

OPTIONS:

- ① All cancers are the same
- ② Skin cancer and brain cancer have different treatment risks and consequences
- ③ Treatment is never invasive
- ④ All patients are identical

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Skin cancer (biopsy, relatively low risk) and brain cancer (major surgery, high risk) have different trade-offs for false positives/negatives. Context determines the appropriate objective function.
- **Why A is wrong:** Different cancers have different risks
- **Why C is wrong:** Some treatments ARE highly invasive
- **Why D is wrong:** Patients have different conditions

EXAM TRAP: Context determines **appropriate trade-offs** - no one-size-fits-all objective function!

QUESTION: How can benchmark datasets introduce bias?

OPTIONS:

- ① They are always representative
- ② They may not be representative of the intended use population
- ③ They always eliminate bias
- ④ They never affect model performance

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Public benchmarks may not reflect real-world populations. Good performance on a benchmark doesn't guarantee good real-world performance and can mask biases that appear in deployment.
- **Why A is wrong:** Benchmarks may NOT be representative
- **Why C is wrong:** Benchmarks can CONTAIN bias
- **Why D is wrong:** Benchmarks significantly affect perceived performance

EXAM TRAP: Benchmark datasets may **not reflect real-world populations** - test in the intended context!

QUESTION: How can deployment context introduce bias?

OPTIONS:

- ① Algorithms always perform equally in all contexts
- ② Algorithms used outside their designed context can lead to performance loss
- ③ Context never matters
- ④ All contexts are identical

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Algorithms designed for one context may not work in another (e.g., recidivism prediction for sentence length). Historical data may be outdated, and user values may differ from developers.
- **Why A is wrong:** Performance varies by context
- **Why C is wrong:** Context significantly matters
- **Why D is wrong:** Contexts vary widely

EXAM TRAP: Deployment context = **critical for performance and fairness** - validate in intended context!

QUESTION: Why is human oversight important for AI systems?

OPTIONS:

- ❶ Humans are never needed
- ❷ Ensuring safe and fair algorithms requires human oversight throughout
- ❸ AI systems are always perfectly safe
- ❹ Oversight is optional

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Human oversight is essential throughout the AI lifecycle - data collection, model development, deployment, and regular auditing. This is critical in consequential domains.
- **Why A is wrong:** Human oversight is ESSENTIAL
- **Why C is wrong:** AI systems are NOT always perfectly safe
- **Why D is wrong:** Oversight is CRITICAL, not optional

EXAM TRAP: Human oversight is [essential throughout the AI lifecycle](#) - not optional!

QUESTION: Which is NOT a source of algorithmic unfairness?

OPTIONS:

- ① Problem specification and feature selection
- ② Data collection and composition
- ③ Model development
- ④ Always using perfect data

ANSWER: D

DETAILED EXPLANATION:

- **Why D is correct:** Perfect data does NOT exist - it's a hypothetical, not a source. All real data has limitations and potential biases.
- **Why A is wrong:** This IS a source of unfairness
- **Why B is wrong:** This IS a source of unfairness
- **Why C is wrong:** This IS a source of unfairness

EXAM TRAP: Perfect data **does NOT exist** - all data has limitations that can introduce bias!

QUESTION: What is explainability in AI?

OPTIONS:

- ① The ability to make predictions
- ② The capacity to explain in understandable terms how an AI system makes decisions
- ③ The speed of the algorithm
- ④ The accuracy of the algorithm

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Explainability is the capacity to explain decisions in understandable terms, often after the fact (ex-post). It is essential for accountability and trust in high-stakes domains.
- **Why A is wrong:** This is prediction capability
- **Why C is wrong:** Speed is different from explainability
- **Why D is wrong:** Accuracy is different from explainability

EXAM TRAP: Explainability = explaining decisions after the fact - essential for accountability!

QUESTION: What is interpretability in AI?

OPTIONS:

- ❶ Explaining decisions after the fact
- ❷ The degree to which a human can comprehend and predict the model's behavior inherently
- ❸ The speed of the algorithm
- ❹ The accuracy of the algorithm

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Interpretability is the degree to which a human can understand the model's behavior through built-in mechanisms (ex-ante). Inherently interpretable models like decision trees can be understood from the start.
- **Why A is wrong:** This is explainability
- **Why C is wrong:** Speed is different
- **Why D is wrong:** Accuracy is different

EXAM TRAP: Interpretability = **inherently understandable models** - built-in, not after the fact!

QUESTION: What is transparency in AI?

OPTIONS:

- ❶ Making predictions quickly
- ❷ The openness and accessibility of an AI system's design, algorithms, data, and decision-making processes
- ❸ The accuracy of the algorithm
- ❹ The speed of the algorithm

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Transparency is the openness and accessibility of an AI system's design, algorithms, data, and processes. It builds trust, enables oversight, and facilitates collaboration.
- **Why A is wrong:** Speed is different
- **Why C is wrong:** Accuracy is different
- **Why D is wrong:** Speed is different

EXAM TRAP: Transparency = **openness and accessibility** - builds trust and enables oversight!

QUESTION: What is the black-box problem in AI?

OPTIONS:

- ① Models that are always transparent
- ② Models that are too complex to be interpreted by humans
- ③ Models that are always simple
- ④ Models that are always accurate

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The black-box problem occurs when models (especially neural networks) are too complex for humans to understand their intrinsic architecture. This creates explainability and accountability issues.
- **Why A is wrong:** Black boxes are NOT transparent
- **Why C is wrong:** Black boxes are COMPLEX
- **Why D is wrong:** Black boxes may NOT be accurate

EXAM TRAP: Black-box problem = **models too complex to interpret** - creates accountability gaps!

QUESTION: What are feature importance scores used for?

OPTIONS:

- ① To make predictions faster
- ② To rank the importance of input features for outcomes
- ③ To reduce the number of data points
- ④ To increase model complexity

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Feature importance scores rank the importance of input features for outcomes. Used in XAI, they reveal which factors have most influence and help identify potential biases.
- **Why A is wrong:** Feature importance is about interpretation
- **Why C is wrong:** It doesn't reduce data points
- **Why D is wrong:** It explains, not increases complexity

EXAM TRAP: Feature importance = **ranks feature influence** - helps identify biases!

QUESTION: What do Shapley values calculate?

OPTIONS:

- ❶ The accuracy of a model
- ❷ The contribution of each input feature to the model's output
- ❸ The speed of the model
- ❹ The size of the model

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Shapley values calculate the contribution of each input feature by measuring the marginal effect of including each feature across all possible combinations. Based on game theory, they provide fair attribution.
- **Why A is wrong:** Accuracy is different
- **Why C is wrong:** Speed is different
- **Why D is wrong:** Size is different

EXAM TRAP: Shapley values = [feature contribution calculation](#) - based on game theory!

QUESTION: What does LUCID do in XAI?

OPTIONS:

- ① Makes predictions faster
- ② Works backward from algorithmic output to reveal internal logic
- ③ Increases model complexity
- ④ Removes features

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** LUCID (Locating Unfairness through Canonical Inverse Design) works backward from a particular output to create a distribution over the input space, revealing the model's internal logic and locating potential unfairness.
- **Why A is wrong:** Speed is not LUCID's purpose
- **Why C is wrong:** LUCID helps understand complexity
- **Why D is wrong:** LUCID doesn't remove features

EXAM TRAP: LUCID works **backward from output to reveal logic** - locates unfairness!

QUESTION: What are surrogate models?

OPTIONS:

- ① More complex models that replace simpler ones
- ② Simpler, more interpretable models that approximate complex AI systems
- ③ Models that are always less accurate
- ④ Models that cannot be interpreted

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Surrogate models are simpler, more interpretable models (like linear regression or decision trees) that approximate the behavior of complex AI systems to enhance interpretability.
- **Why A is wrong:** Surrogate models are SIMPLER
- **Why C is wrong:** Surrogate models can be quite accurate locally
- **Why D is wrong:** Surrogate models are INTERPRETABLE

EXAM TRAP: Surrogate models are [simpler, interpretable approximations](#) - used for explanation!

QUESTION: What is LIME in XAI?

OPTIONS:

- ① Local Interpretable Model-agnostic Explanations
- ② A method to make models faster
- ③ A technique to increase accuracy
- ④ A method to remove bias

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** LIME is Local Interpretable Model-agnostic Explanations. It approximates a complex model's predictions locally with an interpretable model, providing explanations that are specific to individual predictions.
- **Why B is wrong:** LIME is about explanation, not speed
- **Why C is wrong:** LIME is about interpretability, not accuracy
- **Why D is wrong:** LIME helps identify bias, doesn't remove it directly

EXAM TRAP: LIME = Local Interpretable Model-agnostic Explanations - explains individual predictions!

QUESTION: What is a challenge with XAI techniques?

OPTIONS:

- ① They always produce perfect explanations
- ② Oversimplification can distort understanding
- ③ They are never computationally expensive
- ④ They always eliminate bias

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Oversimplification of complex decisions can distort our understanding and lead to misinterpretation. Additionally, XAI techniques can be computationally expensive and are ex-post.
- **Why A is wrong:** Explanations can be imperfect
- **Why C is wrong:** XAI CAN be computationally expensive
- **Why D is wrong:** XAI helps identify, not eliminate bias

EXAM TRAP: XAI can **oversimplify and distort understanding** - a significant challenge!

QUESTION: What is secrecy as a form of opaqueness?

OPTIONS:

- ① When algorithms are fully transparent
- ② When individuals are not aware of being subject to algorithmic decision making
- ③ When algorithms are always explained
- ④ When all data is public

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Secrecy occurs when individuals are not aware of being subject to algorithmic profiling. Data is extracted without their knowledge (e.g., social media data used for targeted advertising).
- **Why A is wrong:** Secrecy is the OPPOSITE of transparent
- **Why C is wrong:** Secrecy means NOT explained
- **Why D is wrong:** Secrecy means NOT public

EXAM TRAP: Secrecy = **unawareness of algorithmic decision making** - violates transparency and consent!

QUESTION: What is confidentiality as a form of opaqueness?

OPTIONS:

- ① When algorithms are completely transparent
- ② When the algorithmic decision-making process is opaque to affected parties
- ③ When all data is public
- ④ When algorithms are always explained

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Confidentiality occurs when affected parties know they're subject to AI but don't have knowledge of how decisions are made. This is a main concern in the AI discourse.
- **Why A is wrong:** Confidentiality means NOT transparent
- **Why C is wrong:** Confidentiality means NOT public
- **Why D is wrong:** Confidentiality means NOT explained

EXAM TRAP: Confidentiality = **opaque decision-making process** - can't understand why decisions were made!

QUESTION: What is specialized knowledge as a form of opacity?

OPTIONS:

- ① When everyone can understand algorithms
- ② When understanding algorithms requires specialized training
- ③ When algorithms are always simple
- ④ When no knowledge is needed

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Specialized knowledge opacity occurs when even with access to source code and data, understanding requires specialized training. This creates unequal access to understanding.
- **Why A is wrong:** Specialized knowledge means NOT everyone understands
- **Why C is wrong:** Algorithms may be complex
- **Why D is wrong:** Knowledge IS needed

EXAM TRAP: Specialized knowledge = **requires training to understand** - not unique to AI!

QUESTION: What is a valid reason for algorithmic opacity?

OPTIONS:

- ① To hide discrimination
- ② Security (spam filters) and proprietary interests
- ③ To avoid accountability
- ④ To confuse users

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Security (making code public enables circumvention) and proprietary interests (trade secrets) are legitimate reasons for some opacity. Must balance transparency with these interests.
- **Why A is wrong:** Hiding discrimination is NOT valid
- **Why C is wrong:** Avoiding accountability is NOT valid
- **Why D is wrong:** Confusing users is NOT valid

EXAM TRAP: Legitimate reasons include [security and proprietary interests](#) - not hiding discrimination!

QUESTION: What is the most comprehensive statement about explainability?

OPTIONS:

- ❶ Explainability is always easy to achieve
- ❷ Explainability, interpretability, and transparency are essential for accountability and trust in AI systems
- ❸ Explainability is never needed
- ❹ Only black-box models are used

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Explainability (ex-post understanding), interpretability (inherent understanding), and transparency (openness) are all essential for accountability and trust in AI systems, especially in high-stakes domains.
- **Why A is wrong:** Explainability can be challenging
- **Why C is wrong:** Explainability IS needed for accountability
- **Why D is wrong:** Many models are interpretable

EXAM TRAP: All three are **essential for accountability and trust** - not optional!

QUESTION: What is human autonomy in the context of AI?

OPTIONS:

- ① The ability to make any decision
- ② The ability to hold beliefs, make, and execute decisions of our own
- ③ The ability to control AI systems
- ④ The ability to avoid all technology

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Human autonomy has two dimensions: internal (values, beliefs, decisions that are our own) and external (freedom and opportunity to execute decisions). It requires meaningful human oversight.
- **Why A is wrong:** Autonomy is more than making any decision
- **Why C is wrong:** This is control, not autonomy
- **Why D is wrong:** Avoiding technology is not autonomy

EXAM TRAP: Autonomy = own beliefs and decisions + ability to execute - not just control!

QUESTION: What did the Cambridge Analytica scandal demonstrate?

OPTIONS:

- ❶ The benefits of data collection
- ❷ How big data could be used to manipulate voter behavior
- ❸ That data privacy is not important
- ❹ That AI never affects elections

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Cambridge Analytica collected Facebook user data, inferred personality profiles from online behavior, and used this for psychologically tailored political advertising to manipulate voter behavior.
- **Why A is wrong:** The scandal showed MISUSE of data
- **Why C is wrong:** The scandal highlighted privacy concerns
- **Why D is wrong:** AI CAN affect elections

EXAM TRAP: Cambridge Analytica = **political manipulation through data** - powerful demonstration of AI misuse!

QUESTION: What did the Facebook emotional contagion experiment show?

OPTIONS:

- ❶ That emotions cannot be transmitted online
- ❷ That recommendation algorithms can shape and manipulate user experience
- ❸ That users always consent to experiments
- ❹ That AI has no effect on emotions

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Researchers changed users' timelines to display predominantly positive or negative posts, and users' own posts became more positive/negative. This demonstrated the power of algorithms to shape user experience.
- **Why A is wrong:** Emotions CAN be transmitted online
- **Why C is wrong:** Users did NOT consent
- **Why D is wrong:** AI CAN affect emotions

EXAM TRAP: Experiment showed **AI can manipulate emotional state** - without user consent!

QUESTION: How can AI manipulation be prevented?

OPTIONS:

- ① By ignoring all risks
- ② Through human oversight and regulatory oversight
- ③ By eliminating all AI
- ④ By never collecting data

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Prevention requires human oversight throughout development and deployment, regulatory oversight (e.g., EU AI Act), transparency obligations, and balancing benefits with risks.
- **Why A is wrong:** Ignoring risks is not prevention
- **Why C is wrong:** Eliminating AI is unrealistic
- **Why D is wrong:** Some data collection is necessary

EXAM TRAP: Prevention requires **human and regulatory oversight** - not ignoring or eliminating!

QUESTION: What is the most comprehensive statement about autonomy and manipulation?

OPTIONS:

- ❶ AI never affects autonomy
- ❷ AI systems can shape our experience and potentially manipulate us, requiring oversight and regulation
- ❸ Manipulation is always beneficial
- ❹ Autonomy is not important

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** AI shapes online experience and can influence behavior through algorithms (Cambridge Analytica, emotional contagion). This requires oversight and regulation to prevent manipulation.
- **Why A is wrong:** AI CAN affect autonomy
- **Why C is wrong:** Manipulation is generally harmful
- **Why D is wrong:** Autonomy IS important

EXAM TRAP: AI can **shape experience and manipulate** - requires robust oversight!

QUESTION: What did the Uber driverless vehicle crash demonstrate?

OPTIONS:

- ❶ That autonomous vehicles are always safe
- ❷ The risks of over-dependence on AI in critical systems
- ❸ That safety mechanisms are never needed
- ❹ That AI never fails

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The crash demonstrated multiple failures: vehicle systems failed to recognize the pedestrian, safety mechanisms were deactivated, and the human safety driver was distracted (automation bias). This shows risks of over-reliance on AI.
- **Why A is wrong:** The crash showed they are NOT always safe
- **Why C is wrong:** Safety mechanisms ARE needed
- **Why D is wrong:** AI CAN fail

EXAM TRAP: Uber crash = risks of over-reliance on AI - demonstrates multiple failure points!

QUESTION: What is automation bias?

OPTIONS:

- ① The tendency to distrust automated systems
- ② The tendency of humans to over-rely on automated systems
- ③ The tendency to always use manual methods
- ④ The tendency to ignore technology

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Automation bias is the tendency of humans to over-rely on automated systems, leading to complacency, reduced vigilance, and errors of oversight. Particularly problematic in critical situations.
- **Why A is wrong:** This is distrust, not over-reliance
- **Why C is wrong:** Automation bias is about using automation
- **Why D is wrong:** Automation bias is about over-relying

EXAM TRAP: Automation bias = **over-reliance on automated systems** - leads to complacency!

QUESTION: What is skill atrophy in the context of AI?

OPTIONS:

- ❶ Improving skills through AI use
- ❷ Decay in essential skills from prolonged reliance on automation
- ❸ Learning new skills quickly
- ❹ Maintaining all skills

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Skill atrophy occurs when operators move from active decision makers to passive observers, causing essential skills to decay. This is a consequence of automation bias and prolonged automation use.
- **Why A is wrong:** This is skill improvement
- **Why C is wrong:** Skill atrophy is about DECAY, not learning
- **Why D is wrong:** Skill atrophy is about LOSING skills

EXAM TRAP: Skill atrophy = [decay from over-reliance on automation](#) - loss of expertise!

QUESTION: How can automation bias be mitigated?

OPTIONS:

- ① By removing all automation
- ② By informing users and requiring periodic human input
- ③ By ignoring the problem
- ④ By increasing automation

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Mitigation requires informing users about the risks and limitations of AI, and implementing design mechanisms that require periodic human input and verification (e.g., eye tracking with alerts).
- **Why A is wrong:** Removing all automation is unrealistic
- **Why C is wrong:** Ignoring the problem is not mitigation
- **Why D is wrong:** Increasing automation worsens the problem

EXAM TRAP: Mitigation = **user education** + **periodic human verification** - keep operators engaged!

QUESTION: What is a concern with AI-driven mental health support?

OPTIONS:

- ① AI systems have perfect empathy
- ② AI systems have no empathy, which is crucial for mental health support
- ③ AI systems are always better than humans
- ④ AI systems never fail in mental health

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** While AI can give seemingly empathetic responses, it has no genuine empathy - which is crucial for mental health support. Positive effects dwindle when patients know responses are AI-generated.
- **Why A is wrong:** AI does NOT have perfect empathy
- **Why C is wrong:** AI is NOT always better than humans
- **Why D is wrong:** AI CAN fail in mental health

EXAM TRAP: AI lacks **genuine empathy** - critical for mental health support!

QUESTION: What are the three main causes of AI-related reputational damage?

OPTIONS:

- ① Speed, cost, and accuracy
- ② Privacy breaches, algorithmic bias, and lack of explainability
- ③ Size, complexity, and speed
- ④ Training data, model type, and deployment

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Research shows the three main causes are privacy breaches (Cambridge Analytica), algorithmic bias (Dutch child benefits scandal), and lack of explainability (customers denied without reasons).
- **Why A is wrong:** Speed, cost, accuracy are not the main causes
- **Why C is wrong:** Size, complexity, speed are not the main causes
- **Why D is wrong:** Training data, model type, deployment are not the main causes

EXAM TRAP: Three main causes = **privacy, bias, explainability** - critical to manage!

QUESTION: What happened in the Dutch child benefits scandal?

OPTIONS:

- ① An algorithm correctly identified fraud
- ② A fraud-detection algorithm falsely denied benefits to thousands of citizens
- ③ All claims were approved
- ④ The algorithm worked perfectly

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** A fraud-detection algorithm deployed by the Dutch government falsely denied and reclaimed child benefits from thousands of citizens, with dire consequences. The scandal led to the Dutch prime minister's resignation.
- **Why A is wrong:** The algorithm incorrectly identified fraud
- **Why C is wrong:** Many claims were DENIED
- **Why D is wrong:** The algorithm failed catastrophically

EXAM TRAP: Dutch scandal = [algorithmic bias with severe consequences](#) - prime minister resigned!

QUESTION: What is competence criticism of companies?

OPTIONS:

- ① Criticism that the company lacks relevant expertise
- ② Criticism that the company has misaligned values
- ③ Criticism that the company is too successful
- ④ Criticism that the company is too transparent

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** Competence criticism arises when there is a perception that the company lacks the relevant expertise and capacities to ensure their algorithms are safe, fair, and explainable.
- **Why B is wrong:** This is value alignment criticism
- **Why C is wrong:** This is not criticism
- **Why D is wrong:** Transparency is generally positive

EXAM TRAP: Competence = **lack of expertise or capacity** - can be fixed through improvement!

QUESTION: What is value alignment criticism of companies?

OPTIONS:

- ❶ Criticism that the company lacks relevant expertise
- ❷ Criticism that the company has misaligned values or acted despite knowing risks
- ❸ Criticism that the company is too successful
- ❹ Criticism that the company is too transparent

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Value alignment criticism occurs when the company was fully aware of the risks but nevertheless proceeded (e.g., Facebook emotional contagion experiment). It suggests a violation of user trust.
- **Why A is wrong:** This is competence criticism
- **Why C is wrong:** This is not criticism
- **Why D is wrong:** Transparency is generally positive

EXAM TRAP: Value alignment = **misaligned values or knowingly proceeding with risks** - harder to fix!

QUESTION: What is **NOT** a reputational risk mitigation strategy?

OPTIONS:

- ① Continuous monitoring and evaluation
- ② Ignoring incidents and hoping they go away
- ③ Ethical AI frameworks
- ④ Stakeholder engagement and communication

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Ignoring incidents is **NOT** a valid mitigation strategy. Effective strategies include continuous monitoring, ethical AI frameworks, stakeholder engagement, and crisis management plans.
- **Why A is wrong:** This **IS** a mitigation strategy
- **Why C is wrong:** This **IS** a mitigation strategy
- **Why D is wrong:** This **IS** a mitigation strategy

EXAM TRAP: Ignoring incidents is **NOT** a valid mitigation strategy - proactive management is essential!

Q81: Superintelligence Definition - Tutorial

QUESTION: What is superintelligence?

OPTIONS:

- ① An AI system that is faster than humans
- ② An AI system that operates beyond human-level intelligence
- ③ An AI system that is less intelligent than humans
- ④ An AI system that only performs one task

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Superintelligence refers to AI that operates beyond human-level intelligence. Such a system could act in ways that threaten humanity, especially if its values are misaligned with human values.
- **Why A is wrong:** Speed is different from intelligence level
- **Why C is wrong:** Superintelligence is ABOVE human level
- **Why D is wrong:** Superintelligence would be general, not narrow

EXAM TRAP: Superintelligence = **beyond human-level intelligence** - central to existential risk!

QUESTION: What is AI alignment?

OPTIONS:

- ① Aligning AI systems with human values
- ② Aligning AI systems with maximum efficiency
- ③ Aligning AI systems with profit maximization
- ④ Aligning AI systems with speed

ANSWER: A

DETAILED EXPLANATION:

- **Why A is correct:** AI alignment is the challenge of ensuring that AI systems' means and goals are aligned with human values. Misaligned AI could cause catastrophic harm even with good intentions.
- **Why B is wrong:** Efficiency is not alignment
- **Why C is wrong:** Profit is not alignment
- **Why D is wrong:** Speed is not alignment

EXAM TRAP: AI alignment = **matching AI to human values** - critical for safe AI!

QUESTION: What is the AI arms race concern?

OPTIONS:

- ① Competing nations developing AI for peaceful purposes
- ② Risks from different stakeholders creating powerful AI without adequate safety measures
- ③ A race to create the fastest AI
- ④ A race to create the most profitable AI

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The AI arms race concern is that competing stakeholders may create powerful AI systems without adequate safety measures, potentially leading to catastrophic conflicts.
- **Why A is wrong:** The concern is about UNSAFE development
- **Why C is wrong:** Speed is not the primary concern
- **Why D is wrong:** Profit is not the primary concern

EXAM TRAP: AI arms race = **unsafe AI development competition** - could lead to catastrophic outcomes!

QUESTION: What is a major economic risk from AI?

OPTIONS:

- ① AI always creates more jobs
- ② AI never affects employment
- ③ Displacement of jobs by AI and widening inequality
- ④ AI has no economic impact

ANSWER: C

DETAILED EXPLANATION:

- **Why C is correct:** AI could displace many jobs (high-skill jobs like accounting are also at risk), creating a skills gap and widening inequality. Predictions vary but Goldman Sachs suggests 1/4 of jobs could be automated.
- **Why A is wrong:** AI may DESTROY some jobs
- **Why B is wrong:** AI DOES affect employment
- **Why D is wrong:** AI HAS significant economic impact

EXAM TRAP: Economic risks = **job displacement and inequality** - requires workforce re-skilling!

QUESTION: How could AI exacerbate global inequality?

OPTIONS:

- ❶ AI is evenly distributed globally
- ❷ Nations at the center of AI development leap ahead, leaving others behind
- ❸ AI has no effect on global inequality
- ❹ AI reduces inequality automatically

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Most AI firms are based in the US, and technological adoption varies widely. Nations at the center of AI development could leap ahead in economic growth and influence, leaving others behind.
- **Why A is wrong:** AI is NOT evenly distributed
- **Why C is wrong:** AI CAN exacerbate inequality
- **Why D is wrong:** AI does NOT automatically reduce inequality

EXAM TRAP: AI could widen global inequality - creates geopolitical tensions!

QUESTION: According to McKinsey (2025), what percentage of organizations use GenAI?

OPTIONS:

- 1 33%
- 2 71%
- 3 50%
- 4 95%

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** McKinsey 2025 found that 71% of organizations regularly use GenAI, up from 33% in 2023. Adoption is particularly high in digitized industries like banking (58% embrace GenAI).
- **Why A is wrong:** This was the 2023 figure
- **Why C is wrong:** The figure is higher than 50%
- **Why D is wrong:** This is the BCG figure for professional services

EXAM TRAP: GenAI adoption = 71% of organizations (2025 McKinsey) - rapid growth!

QUESTION: What is hallucination in GenAI?

OPTIONS:

- ① Always correct outputs
- ② Factually incorrect, misleading, or fabricated outputs that appear credible
- ③ Outputs that are never credible
- ④ Outputs that are always correct

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Hallucinations are factually incorrect or misleading outputs that appear credible to non-experts. They are particularly problematic in legal, financial, and healthcare contexts.
- **Why A is wrong:** Hallucinations are INCORRECT
- **Why C is wrong:** Hallucinations APPEAR credible
- **Why D is wrong:** Hallucinations are NOT correct

EXAM TRAP: Hallucination = **incorrect but credible outputs** - hard for non-experts to detect!

QUESTION: What happened in the Air Canada chatbot case?

OPTIONS:

- ❶ The chatbot worked perfectly
- ❷ The chatbot offered a discount that shouldn't have been available, and the company was held responsible
- ❸ The chatbot was a separate legal entity
- ❹ The company won the case

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** The chatbot offered a discount that shouldn't have been available. Air Canada argued the chatbot was a "separate legal entity," but the tribunal rejected this, ruling the company was responsible.
- **Why A is wrong:** The chatbot made an error
- **Why C is wrong:** The argument was REJECTED
- **Why D is wrong:** Air Canada LOST the case

EXAM TRAP: Companies are **liable for chatbot outputs** - can't claim ignorance!

QUESTION: Why is unpredictability a risk in GenAI?

OPTIONS:

- ① GenAI outputs are always predictable
- ② Unpredictability undermines standardization across users and time
- ③ Unpredictability is always beneficial
- ④ Unpredictability never affects compliance

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** GenAI systems are unpredictable in their outputs, which undermines efforts to standardize outputs across users and time. This poses challenges for auditability and compliance.
- **Why A is wrong:** GenAI outputs are NOT always predictable
- **Why C is wrong:** Unpredictability is generally problematic
- **Why D is wrong:** Unpredictability CAN affect compliance

EXAM TRAP: Unpredictability undermines **standardization and auditability** - compliance risk!

QUESTION: What is **prompt injection**?

OPTIONS:

- ❶ A technique to improve outputs
- ❷ Hidden instructions within user inputs to cause unintended behavior
- ❸ A method to increase accuracy
- ❹ A technique to remove bias

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Prompt injection involves hidden instructions in user inputs designed to cause unintended behavior, manipulating the AI system's response.
- **Why A is wrong:** Prompt injection causes PROBLEMS
- **Why C is wrong:** Prompt injection is about manipulation
- **Why D is wrong:** Prompt injection doesn't remove bias

EXAM TRAP: Prompt injection = **hidden instructions to cause unintended behavior** - security risk!

QUESTION: What is jailbreaking in the context of GenAI?

OPTIONS:

- ❶ Making the model faster
- ❷ Bypassing a model's safety mechanisms through crafted prompts
- ❸ A method to improve outputs
- ❹ A technique to increase accuracy

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Jailbreaking involves crafted prompts that bypass a model's safety mechanisms, overriding safety or policy constraints to cause unintended or unsafe behavior.
- **Why A is wrong:** Speed is not the purpose
- **Why C is wrong:** Jailbreaking is about bypassing safety
- **Why D is wrong:** Jailbreaking does NOT increase accuracy

EXAM TRAP: Jailbreaking = [bypassing safety mechanisms](#) - security and safety risk!

QUESTION: How can GenAI cause data privacy violations?

OPTIONS:

- ❶ GenAI never accesses personal data
- ❷ Prompts containing PII may be inadvertently stored, shared, or used for training
- ❸ Data privacy is not a concern
- ❹ GenAI always protects privacy

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** User prompts may contain personally identifiable information (PII) that can be inadvertently stored, shared, or used for model training. Researchers have extracted emails and phone numbers from LLMs.
- **Why A is wrong:** GenAI CAN access personal data
- **Why C is wrong:** Data privacy IS a concern
- **Why D is wrong:** GenAI does NOT always protect privacy

EXAM TRAP: GenAI prompts may inadvertently expose PII - privacy risk!

QUESTION: What copyright concern exists with GenAI?

OPTIONS:

- ❶ GenAI never uses copyrighted material
- ❷ GenAI models are trained on web-scraped content including copyrighted material
- ❸ Copyright laws don't apply to AI
- ❹ All GenAI content is public domain

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** GenAI models are trained on vast web-scraped content that includes copyrighted books, images, and articles. This has created legal proceedings (e.g., Getty Images vs. Stability AI).
- **Why A is wrong:** GenAI DOES use copyrighted material
- **Why C is wrong:** Copyright laws DO apply
- **Why D is wrong:** GenAI content is NOT public domain

EXAM TRAP: GenAI training may [infringe copyright](#) - legal and reputational risk!

QUESTION: What are responsibility gaps in GenAI adoption?

OPTIONS:

- ① Clear lines of accountability always exist
- ② GenAI tools may be adopted across teams without clear responsibility for oversight
- ③ Responsibility is never an issue
- ④ All employees are always responsible

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** GenAI tools are intuitive and adopted across teams (including non-technical teams like HR), often without clear responsibility for oversight, quality assurance, and risk mitigation.
- **Why A is wrong:** Responsibility gaps ARE a problem
- **Why C is wrong:** Responsibility IS an issue
- **Why D is wrong:** Not all employees have clear responsibility

EXAM TRAP: GenAI adoption creates **responsibility gaps** across teams - need clear accountability!

QUESTION: What misalignment exists in GenAI adoption?

OPTIONS:

- ① Companies always prioritize safety over speed
- ② Pressure to innovate can outweigh cautionary approaches
- ③ Regulation always prevents problems
- ④ Incentives are always aligned with safety

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Pressure to cut costs and improve productivity can outweigh safety considerations. Teams may be rewarded for cost savings through automation, not for flagging risks or ensuring safe deployment.
- **Why A is wrong:** Safety is NOT always prioritized
- **Why C is wrong:** Regulation is not always effective
- **Why D is wrong:** Incentives are NOT always aligned

EXAM TRAP: Innovation pressure can [outweigh safety considerations](#) - misaligned incentives!

QUESTION: What risk arises from lack of GenAI training?

OPTIONS:

- ① Employees always use GenAI correctly
- ② Employees may use GenAI without understanding acceptable use or data protection obligations
- ③ Training is never needed
- ④ GenAI always protects data

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** LLM chatbots are intuitive, so employees integrate them into workflows without guidance. 4.2% of workers pasted sensitive data into ChatGPT, leading companies like JPMorgan to ban it.
- **Why A is wrong:** Employees may NOT use GenAI correctly
- **Why C is wrong:** Training IS needed
- **Why D is wrong:** GenAI does NOT always protect data

EXAM TRAP: Lack of training leads to **unintentional data exposure** - significant risk!

QUESTION: What is vendor lock-in in GenAI?

OPTIONS:

- ① Complete freedom to switch vendors
- ② Dependence on external providers, limiting auditability and leverage
- ③ Vendors never change pricing
- ④ All providers are identical

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Most organizations rely on external providers from a small set of dominant suppliers, creating dependencies and power asymmetries. Providers may change pricing, models, or usage restrictions without warning.
- **Why A is wrong:** Switching is often difficult
- **Why C is wrong:** Vendors CAN change terms
- **Why D is wrong:** Providers differ significantly

EXAM TRAP: Vendor lock-in = **dependence limiting control** - strategic risk!

QUESTION: What risk does synthetic content pose?

OPTIONS:

- ❶ Synthetic content is always beneficial
- ❷ High-quality misleading content can be created at scale, damaging trust and institutions
- ❸ Synthetic content is easy to identify
- ❹ Synthetic content never affects elections

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** GenAI lowers the cost of creating high-quality misleading content (fake news, health misinformation, political content), which can damage trust in information and democratic institutions.
- **Why A is wrong:** Synthetic content CAN be harmful
- **Why C is wrong:** Synthetic content is often difficult to identify
- **Why D is wrong:** Synthetic content CAN affect elections

EXAM TRAP: Synthetic content can **damage trust and institutions** - societal risk!

QUESTION: What risk do deepfakes pose?

OPTIONS:

- ① Deepfakes are always harmless
- ② Audio or video deepfakes can convincingly mimic individuals, causing fraud, defamation, or political manipulation
- ③ Deepfakes are easy to detect
- ④ Deepfakes never cause harm

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** Deepfakes can convincingly mimic public figures or private individuals, causing fraud (CEO persuaded to transfer €220,000), defamation, and political manipulation. Even low-effort impersonations can go viral.
- **Why A is wrong:** Deepfakes CAN cause harm
- **Why C is wrong:** Deepfakes are often difficult to detect
- **Why D is wrong:** Deepfakes CAN cause harm

EXAM TRAP: Deepfakes can **cause fraud, defamation, and manipulation** - serious societal risk!

QUESTION: What is the most comprehensive summary of GenAI risk management?

OPTIONS:

- ❶ GenAI risks are purely technical
- ❷ GenAI risks include functional, data-related, organizational, strategic, and societal risks, requiring integrated oversight and governance
- ❸ GenAI risks are not significant
- ❹ Only hallucinations matter

ANSWER: B

DETAILED EXPLANATION:

- **Why B is correct:** GenAI risks span multiple categories: functional (hallucinations, unpredictability), data-related (privacy, copyright), organizational (responsibility gaps, misaligned incentives), strategic (vendor lock-in, regulatory unpredictability), and societal (disinformation, deepfakes).
- **Why A is wrong:** Risks extend beyond technical
- **Why C is wrong:** GenAI risks ARE significant
- **Why D is wrong:** Multiple risk categories exist

EXAM SUCCESS: Understand **all risk categories** and their interconnections for comprehensive risk management!